Dual-Channel Variational Closure, Covariant Completion, and a Reproducible SPARC Benchmark

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Abstract

We study the dual-channel AQUAL potential $\mathcal{F}_{\rm dual}(x) = \frac{1}{2}x^2 - x + \ln(1+x)$, whose constitutive ratio $\mathcal{F}'(x)/x$ is the "simple" MOND interpolation function $\mu(x) = x/(1+x)$, and carry it through to a relativistic completion with quantitative solar-system and cosmological consequences. Given four structural constraints — constitutive-relation form, Padé[1/1] minimality, the MOND boundary conditions, and dual-channel splitting — $\mathcal{F}_{\rm dual}$ is uniquely determined; μ is the functional inverse of the Bayesian odds ratio and obeys the Fisher identity $\mathcal{F}'(x)^2\mathcal{I}(\mu(x)) = x^3$. The necessity of the Padé constraint is not derived, so the uniqueness is conditional. Specifying the Skordis–Złośnik free function as $\mathcal{F}(\mathcal{K}) = \mathcal{F}_{\rm dual}(\sqrt{\mathcal{K}})$ leaves the Friedmann background unmodified and places the cosmological background on the Newtonian branch, giving a linear screening ratio $\mathcal{F}''/\mathcal{F}' \approx 0.004$: the $\sim 76\times$ structure overproduction of a uniform MOND enhancement falls to 0.4% in linear growth. With Vainshtein suppression the solar-system external-field quadrupole is $Q_2 \approx 4.9\times10^{-29}\,\mathrm{s}^{-2}$, a factor 70 below the Cassini bound at natural $\mathcal{O}(1)$ couplings and with no fine-tuning; the residual $Q_2 \sim 10^{-29}\,\mathrm{s}^{-2}$ is a falsifiable prediction. The completion is ghost-free with $c_T = c$. Across 171 SPARC galaxies we measure $a_0 = (1.116 \pm 0.161)\times10^{-10}\,\mathrm{m\,s^{-2}}$. A pre-registered test on 20 intermediate-redshift galaxies favours constant a_0 over the horizon-tied $a_0(z) = \xi c H(z)$ at 5.9σ; we therefore report the horizon interpretation of a_0 as disfavoured by our own data. The formal core is machine-checked in Lean 4 (18 modules, no `sorry`). Non-linear structure formation, exact PPN parameters, and cluster scales remain open.

Artifacts. ORCID 0009-0001-1303-7190. Repository `github.com/Mega-Therion/Res-Nova`, release `v1.6.2`; DOI 10.5281/zenodo.21969121; verified artifact references `VERIFICATION_RUN_003 / 005 / 006 / 007`.

I. INTRODUCTION AND SCOPE

The asymptotic flatness of galactic rotation curves and the empirical Baryonic Tully–Fisher Relation (BTFR) [?] have motivated extensive investigation into modified gravity, including MOND [?] and AQUAL [?]. Prior work in this research program (PAPER_01, now quarantined) proposed a single-channel arcsinh action whose Euler–Lagrange derivative yields arcsinh(x), not the required rational $\mu(x)$, establishing a correspondence failure.

This v1.6.0 assessment supersedes all prior working papers. It systematically delineates: what is mechanically proved [**P**], what is directly computed [**D**], what is cited from literature [**C**], and what remains open [**O**]. The governing standard is the Sovereign Epistemic Covenant; the authoritative version record is `EPISTEMIC_BOUNDARY_v1.5.0.md`; in any conflict between this manuscript and the ledger, the ledger governs.

II. EPISTEMIC CLASSIFICATION AND REPRODUCIBILITY POLICY

All claims are governed by four explicit tiers:

- **[P] Proved / Machine-Verified:** Mathematical propositions mechanically checked in Lean 4 / Mathlib. Lean establishes only that statements follow from their encoded definitions; physical theory selection is not certified.

- **[D] Direct Empirical / Computed:** Quantities evaluated from raw, SHA-256-authenticated observational data using auditable, version-controlled scripts.

- **[C] Cited External Literature:** Established results from peer-reviewed sources.

- **[O] Open / Quarantined:** Conjectured boundary conditions, unproved physical mechanisms, or claims awaiting future derivation or data.

The complete claim-by-claim audit matrix is maintained in `CLAIM_EVIDENCE_LEDGER.md` and `EPISTEMIC_BOUNDARY_v1.5.0.md`. A human-readable referee index is in `FOR_REFEREES.md`.

III. NO-GO THEOREM FOR THE SINGLE-CHANNEL ACTION (PAPER_01 QUARANTINE)

The single-channel AQUAL potential $\mathcal{F}_{\mathrm{single}}(x) = x \operatorname{arcsinh}(x) - \sqrt{1+x^2}$ yields $\mathcal{F}'(x) = \operatorname{arcsinh}(x)$. This function satisfies $\operatorname{arcsinh}(x) \to 0$ as $x \to 0$ and $\operatorname{arcsinh}(x) \to \infty$ as $x \to \infty$ — the *inverted* MOND correspondence limits. The single-channel branch is therefore correspondence-false and is permanently quarantined as PAPER_01 [**P**]. This result is machine-certified in `05_lean_formalization/DualChannelDerivation.lean`.

IV. DUAL-CHANNEL VARIATIONAL CLOSURE

A. The Dual-Channel Action

Balancing a bulk kinetic flux channel against a horizon relative-entropy dissipation channel [**motivation, not a theorem; see O3**] isolates the unique two-channel potential:

$$\mathcal{F}_{\text{dual}}(x) = \frac{1}{2}x^2 - x + \ln(1+x), \qquad x \equiv \frac{|\nabla\Phi|}{a_0} \geq 0. \tag{1}$$

B. Variational Derivation

Correction to v1.5.0 and earlier

Prior releases of this manuscript printed the constitutive relation as $\mu(x) \equiv \mathcal{F}'_{\text{dual}}(x) = x - 1 + \frac{1}{1+x} = \frac{x}{1+x}$. The middle expression evaluates to $x^2/(1+x)$, not $x/(1+x)$, so that line conflated $\mathcal{F}'$ with μ. The corrected statement is Eqs. (2)–(3) below. Nothing downstream changes: the AQUAL field equation, $\mathcal{F}''$, both asymptotic limits, and the SPARC benchmark all use μ, whose value $x/(1+x)$ is unchanged. The correct relation $\mathcal{F}' = x\mu$ is the one already used in the D2 uniqueness analysis (§V) and encoded in `DualChannelDerivation.lean`.

Differentiating Eq. (1):

$$\mathcal{F}'_{\text{dual}}(x) = x - 1 + \frac{1}{1+x} = \frac{x^2}{1+x}. \quad [\mathbf{P}] \tag{2}$$

In the AQUAL convention the interpolation function is recovered as the constitutive ratio $\mu = \mathcal{F}'(x)/x$, i.e. $\mathcal{F}(x) = \int_0^x t\,\mu(t)\,dt$:

$$\mu(x) = \frac{\mathcal{F}'_{\text{dual}}(x)}{x} = \frac{x}{1+x}. \quad [\mathbf{P}] \tag{3}$$

The asymptotic limits are exact:

$$\lim_{x\to 0} \mu(x) = x + O(x^2) \quad \text{(deep-MOND / BTFR)}, \quad [\mathbf{P}] \tag{4}$$

$$\lim_{x\to \infty} \mu(x) = 1 - \frac{1}{x} + O(x^{-2}) \quad \text{(Newtonian)}, \quad [\mathbf{P}] \tag{5}$$

Both limits and the underlying algebraic identities are machine-certified in Lean 4 (`DualChannelDerivation.lean`, `PPNLimits.lean`). **Scope note:** the Lean module certifies the channel-balance identity $x - \frac{x}{1+x} = \frac{x^2}{1+x}$ and the bounds $\mu(x) < x$, $\mu(x) < 1$; it does not carry out symbolic differentiation of $\mathcal{F}_{\text{dual}}$. The step from Eq. (1) to Eq. (2) is elementary calculus verified by hand and by computer algebra, not by Lean.

C. Kinetic Convexity and Stability

The second derivative:

$$\mathcal{F}''_{\text{dual}}(x) = \frac{x(x+2)}{(1+x)^2} > 0 \quad \forall\, x > 0. \quad [\mathbf{P}] \tag{6}$$

Strict convexity proves absence of tachyonic and gradient instabilities in the quasistatic Poisson field (`RelativisticStability.lean`).

V. CONDITIONAL UNIQUENESS OF THE DUAL-CHANNEL ACTION (D2)

Section IV exhibits an action that works. This section addresses the sharper question a referee will ask: *why this action?* The answer established here is a **conditional** uniqueness theorem, together with an explicit statement of the condition that is not derived.

A. What does not fix the action

Claim [P]: The correspondence constraints alone — $\mu(0) = 0$, $\mu(\infty) = 1$, monotonicity, and smoothness — do *not* determine $\mathcal{F}$. The standard $\mu_{\text{simple}} = x/(1+x)$, $\mu_{\text{std}} = x/\sqrt{1+x^2}$, and an infinite family of interpolations all satisfy them. Any uniqueness claim therefore requires a genuine additional structural principle, stated openly.

B. Padé minimality

Theorem (Padé uniqueness) [P]. $\mu(x) = x/(1+x)$ is the unique Padé [1/1] rational function $\alpha x/(\beta + \gamma x)$ with $\alpha, \beta, \gamma > 0$ satisfying $\mu(0) = 0$, $\mu(\infty) = 1$, and $\mu'(0) = 1$.

Among all rational interpolations meeting the MOND boundary conditions, $x/(1+x)$ carries minimal degree; every alternative requires degree ≥ 2 in numerator or denominator. In this precise sense the "simple" μ is the minimal-complexity rational interpolation, not merely a convenient one.

C. Odds-ratio and Fisher structure

Theorem (odds-ratio inverse) [P]. With $\text{odds}(p) = p/(1-p)$ for $p \in (0,1)$,

$$\mu(\text{odds}(p)) = p, \qquad \text{odds}(\mu(x)) = x. \tag{7}$$

The interpolation function is exactly the inverse of the canonical Bayesian odds map. It therefore induces a bijection between the acceleration ratio $x = a/a_0$ and a probability $p = \mu(x)$, with $p = 1/2$ precisely at $a = a_0$.

Theorem (Fisher identity) [P]. For the Bernoulli distribution at $p = \mu(x)$, $\mathcal{I}(p) = [p(1-p)]^{-1} = (1+x)^2/x$, and since $\mathcal{F}'_{\text{dual}}(x) = x\,\mu(x)$,

$$\mathcal{F}'_{\text{dual}}(x)^2\,\mathcal{I}(\mu(x)) = x^3. \tag{8}$$

The exponent on the right is the deep-MOND scaling of the action itself ($\mathcal{F} \sim x^3/3$ as $x \to 0$). We report Eq. (8) as a structural identity. Its *interpretation* as evidence of an information-geometric origin for MOND is a conjecture [**O**], not a result.

D. The uniqueness theorem and its condition

Theorem (uniqueness of $\mathcal{F}_{\text{dual}}$) [P]. Given

1. constitutive-relation structure, $\mathcal{F}(x) = \int_0^x t\,\mu(t)\,dt$;

2. Padé [1/1] minimality of μ;

3. MOND boundary conditions $\mu(0) = 0$, $\mu(\infty) = 1$, $\mu'(0) = 1$;

4. dual-channel splitting $\mathcal{F} = \frac{1}{2}x^2 - \mathcal{F}_{\text{corr}}$ with Newtonian channel $\frac{1}{2}x^2$;

the potential $\mathcal{F}_{\text{dual}}(x) = \frac{1}{2}x^2 - x + \ln(1+x)$ is uniquely determined.

What this theorem does not claim [O]

The *necessity of constraint 2* is not derived from a deeper principle. Nothing here explains why μ should be rational at all, nor why the minimal Padé order is the physical one. The theorem therefore relocates the functional freedom — from "which interpolation function?" to "which rational order and which boundary conditions?" — rather than eliminating it. Whether Padé minimality follows from an information-theoretic variational principle (minimum Fisher information, or maximum entropy on the Bernoulli manifold) is open. We claim conditional uniqueness only. This target is scored [**P**/**O**] in the ledger, and any reading of this section as a parameter-free first-principles derivation of μ is a misreading.

VI. EFFECTIVE WEAK-FIELD MODEL

In the weak-field, non-relativistic limit, the modified gravitational potential satisfies the AQUAL Poisson equation [**?**]:

$$\nabla \cdot \left[\mu\left(\frac{|\nabla\Phi|}{a_0} \right) \nabla\Phi \right] = 4\pi G \rho_{\rm bar}, \tag{9}$$

with $\mu(x) = x/(1+x)$ [**P**]. Under spherical symmetry this reduces to the algebraic relation $g \cdot \mu(g/a_0) = g_{\rm bar}$, giving the closed-form acceleration:

$$g = g_{\rm bar} \left[\frac{1}{2} + \sqrt{\frac{1}{4} + \frac{a_0}{g_{\rm bar}}} \right]. \tag{10}$$

VII. ACCELERATION-SCALE MEASUREMENT AND OPEN HORIZON IDENTITY

A. Fitted Measurement

A bootstrap-over-galaxies a_0 measurement on 171 SPARC galaxies (3,375 kinematic points, `A0_MEASUREMENT.json`) yields:

$$a_0 = (1.116 \pm 0.128_{\rm stat} \pm 0.097_{\rm syst}) \times 10^{-10}\,{\rm m\,s^{-2}} \quad (14.4\%\ \text{total}). \quad [\mathbf{D}] \tag{11}$$

The horizon anchor $cH_0/(2\pi) \approx 1.044 \times 10^{-10}\,{\rm m\,s^{-2}}$ (Planck $H_0 = 67.4\,{\rm km\,s^{-1}Mpc^{-1}}$) lies 0.46σ away; MOND's canonical 1.2×10^{-10} lies 0.52σ away. These two anchors are not separated at $z = 0$ within current systematics.

Superseded value (do not cite): An earlier in-sample MAP run reported $a_0 = (9.433 \pm 0.050) \times 10^{-11}$. That error bar treated 3,391 radial points as independent and leaked the global a_0 into every cross-validation test fold. It is permanently superseded by Eq. (11).

B. Horizon Identity — Open Problem O1

Thermal equilibrium between the de Sitter Gibbons–Hawking temperature [?] and the Unruh temperature [?] yields $a = cH_0$; the universal KMS 2π factors cancel identically. This cancellation is now formally proved in Lean 4 (`HorizonScale.lean`: `kms_cancellation_equilibrium`, $\xi = 1$ [**P**]). The additional $1/(2\pi)$ divisor in $a_0 = cH_0/(2\pi)$ is therefore an **open boundary normalization** [**O**] (O1), not a first-principles derivation. The 2π divisor must be postulated as an auxiliary hypothesis; no derivation from the thermal equilibrium condition produces it.

C. Pre-Registered Redshift Test — O4 Result

A pre-registered hypothesis test (`PREREG_A0_OF_Z.md`, written before data evaluation) compares two fully-specified models on 20 MUSE-DARK III / HUDF galaxies ($0.41 \leq z \leq 1.44$, median $z = 0.94$):

- H_{const}: $a_0(z) = a_0(0)$ (constant; canonical MOND).
- H_{horizon}: $a_0(z) = \xi \cdot c \cdot H(z)$ with ξ frozen from the $z = 0$ SPARC anchor.

Both models share the same frozen $\mu(x) = x/(1+x)$ and zero NFW parameters. No parameters are re-fitted at the test stage. The pre-registered verdict threshold is $|\Delta\chi^2| \geq 9.0$ for 3σ exclusion.

Result [**D**]: $\chi^2(H_{\mathrm{const}}) = 5.37$, $\chi^2(H_{\mathrm{horizon}}) = 40.49$, $\Delta\chi^2 = -35.1$. The constant-a_0 hypothesis is favoured over the horizon-tied hypothesis at 5.9σ, excluding H_{horizon} at $> 3\sigma$ (`A0_OF_Z_REPORT.json`). The horizon evolution prediction $a_0(z) = \xi c H(z)$ is *not supported* by the current intermediate-redshift kinematic data.

Caveat: The sample is 20 galaxies with one kinematic point each. The test is decisive in its pre-registered form but should be re-run with larger JWST/NIRSpec samples as they become available. The result does not falsify MOND itself — it favours constant a_0 (canonical MOND) over the horizon-tied variant.

VIII. SPARC BENCHMARK: DATA, METHODS, AND MATCHED-PARAMETER COMPARISON

A. Observational Sample

We use the complete Spitzer Photometry and Accurate Rotation Curves (SPARC) database [**?**]: 175 disk galaxies, 3,391 kinematic observations. Raw data SHA-256:

`e76e6752164b...c05f96d095456e067bdd5c6da59d2be4ec70c1eb`

The a_0 measurement sub-sample is 171 galaxies / 3,375 points (4 galaxies excluded for insufficient data coverage).

B. Specification Definitions

- **GOD Tier 0 (Zero-Parameter):** $\Upsilon_{\text{disk}} = \Upsilon_{\text{bulge}} = f_d = 1.0$ fixed; a_0 from horizon formula. Tests the a_0 source hypothesis directly.

- **MOND Tier 0:** Same, with $a_0 = 1.2 \times 10^{-10}\,\mathrm{m\,s^{-2}}$ from literature [**?**].

- **GOD/MOND Tier 1 ($N_{\mathbf{par}} = 374$):** 174 Υ_{disk} + 199 Υ_{bulge} (where applicable) + 1 global a_0 under $\mathcal{N}(0.5, 0.125^2)$ / $\mathcal{N}(0.7, 0.175^2)$ priors.

- **NFW Free-c ($N_{\mathbf{par}} = 716$):** Per-galaxy NFW halo concentration c free. 97/171 railed at $c = 1$; not a fair ΛCDM proxy.

- **NFW Cosmological-c Prior ($N_{\mathbf{par}} = 716$):** Concentration c constrained to cosmological mass–concentration relation. This is the fair ΛCDM-comparable row.

Specification	**Free Params**	**Median $\chi^2_{\mathbf{data}}/N_g$**	**Aggregate**	**Role**
GOD Tier 0 (horizon a_0, fixed M/L) [**D**]	0	9.20	—	Tests a_0 source hypothesis
MOND Tier 0 (literature a_0) [**D**]	0	11.35	—	Zero-param control
GOD Tier 1 [**D**]	**374**	**2.95**	—	**Dual-channel μ, horizon a_0**
MOND Tier 1 [**D**]	374	2.89	—	Dual-channel μ, fitted a_0
NFW Free c [**D**]	716	1.92	—	97/171 railed at $c = 1$; not ΛCDM
NFW Cosmological-c Prior [**D**]	**716**	**5.62**	—	**Fair ΛCDM row; 342 extra params vs GOD**

TABLE I. Matched-parameter SPARC benchmark on 171 galaxies. GOD Tier 1 achieves median $\chi^2/N_g = 2.95$ with 374 parameters; the fair cosmological NFW model requires 342 additional parameters (716 total) for median $\chi^2/N_g = 5.62$. Parameter counts and fit results frozen in `PARAMETER_LEDGER.json` and `NFW_CONSTRAINED.json` (`VERIFICATION_RUN_003`).

C. Matched-Parameter Results

D. Pre-Registered Falsification Contract

Pre-Registered Interpretation Contract

1. **Closure Test (A_4):** If the canonical 374-parameter GOD Tier 1 aggregate χ^2/dof underperforms the legacy μ_{simple} zero-parameter baseline in a like-for-like test, the derived closure is falsified.

2. **Scale Test (F_3):** Any persistent tension in fitted a_0 vs. horizon anchor exceeding 3σ falsifies the horizon-normalization narrative.

3. **No Silent Spec Changes:** Sample, error models, and priors frozen across all runs; changes must be versioned in the ledger.

4. **Parameter Ledger:** All parameter counts reported against identical galaxy samples. NFW comparisons must use cosmological-c priors to constitute a fair test.

IX. 4D COVARIANT COMPLETION: OBSTRUCTIONS AND SKORDIS–ZŁOŚNIK EMBEDDING

A. Pure RAQUAL Superluminality (Falsified)

In static galaxy halos with spacelike gradient ($x > 0$), the perturbation speed parallel to the gradient for any RAQUAL/k-essence action is:

$$c_{\parallel}^2 = 1 + \left.\frac{2y\mathcal{F}''(y)}{\mathcal{F}'(y)}\right|_{y=x^2} = 1 + \frac{1}{1+x} > 1 \quad \forall\, x > 0. \quad [\mathbf{P}] \tag{12}$$

Pure RAQUAL is **falsified** on superluminality grounds. Machine-certified in `CovariantCompletion.lean`.

B. Disformal Completion (Falsified by GW170817)

The disformal metric $\tilde{g}_{\mu\nu} = e^{-2\phi} g_{\mu\nu} - 2\sinh(2\phi) u_\mu u_\nu$ yields photon speed $c_\gamma = e^{2\phi}$ on $\tilde{g}$ and graviton speed $c_T = 1$ on g. The speed ratio:

$$\frac{c_T}{c_\gamma} = e^{-2\phi} = 1 \iff \phi = 0. \tag{13}$$

For deep-MOND halos ($\phi \sim 10^{-6}$) or cosmological backgrounds ($\phi \sim 0.1$), $|e^{-2\phi} - 1| \gg 10^{-15}$, violating the GW170817/GRB 170817A multi-messenger bound by 9–14 orders of magnitude. The disformal completion $B(\phi) = -2\sinh(2\phi)$ is **falsified** [**P**]. Machine-certified in `TensorSpeed.lean`.

C. Skordis–Złośnik Embedding (Viable, Proved)

The Skordis–Złośnik (2021) [**?**] action admits a free kinetic function $\mathcal{J}(\mathcal{Y})$ on the spacelike projected gradient $\mathcal{Y} \equiv g^{\mu\nu} \hat{\nabla}_\mu \phi \, \hat{\nabla}_\nu \phi / a_0^2$. Setting:

$$\mathcal{J}(\mathcal{Y}) = \frac{1}{2}\mathcal{Y} - \sqrt{\mathcal{Y}} + \ln(1 + \sqrt{\mathcal{Y}}) \equiv \mathcal{F}_{\mathrm{dual}}(\sqrt{\mathcal{Y}}) \tag{14}$$

yields $2\mathcal{J}'(\mathcal{Y}) = \sqrt{\mathcal{Y}}/(1 + \sqrt{\mathcal{Y}}) = \mu(\sqrt{\mathcal{Y}})$ [**P**].

Consequences (all machine-certified in `SkordisZlosnikEmbedding.lean`, `TensorSpeed.lean`):

- **GW170817:** Matter and photons couple to $g_{\mu\nu}$ directly; $c_T = c_\gamma = c$, so $|c_T/c_\gamma - 1| = 0 \leq 10^{-15}$ **identically** [**P**].
- **Lensing / PPN:** Dynamical vector A^μ shear cancels scalar trace discrepancies, maintaining $\Phi = \Psi$ ($\gamma_{\mathrm{PPN}} = 1$, proved in `CovariantCompletion.lean`) without disformal distortion [**P**].
- **Preferred-frame bound:** $\alpha_1 = -2K$; solar system constraint $|\alpha_1| \lesssim 10^{-4}$ places $|K| \lesssim 5 \times 10^{-5}$ [**P**]/[**D**].
- **Ghost freedom:** $\mathcal{J}''(\mathcal{Y}) = \frac{1}{4\sqrt{\mathcal{Y}}(1+\sqrt{\mathcal{Y}})^2} > 0$ [**P**].
- **Cosmological decoupling:** On FLRW, $\hat{\nabla}_\mu \phi = 0 \implies \rho_\phi = 0$; the MOND kinetic sector contributes zero dynamical dark energy [**P**].

X. RMOND COMPLETION WITH THE DUAL-CHANNEL FREE FUNCTION (D7)

Section IX establishes that the dual-channel action *can* be embedded in the Skordis–Złośnik framework. This section specifies the embedding as a complete relativistic theory and derives its screening behaviour, which is what makes the solar-system (§XI) and cosmological (§XII) sectors viable.

A. The action

We adopt the Skordis–Złośnik action [**?**]

$$S = \int d^4x\sqrt{-g}\left[\frac{R}{16\pi G} + \frac{a_0^2}{16\pi G}\mathcal{F}(\mathcal{K}) + \mathcal{L}_\phi + \lambda(A^\mu A_\mu + 1) + \mathcal{L}_{\mathrm{m}}[\tilde{g}_{\mu\nu}]\right], \tag{15}$$

with A^μ a unit-timelike vector, $\tilde{g}_{\mu\nu} = e^{2\phi}g_{\mu\nu} + \sigma^2 A_\mu A_\nu$ the physical metric, and the Einstein-aether kinetic scalar

$$\mathcal{K} = c_1(\nabla_\mu A_\nu)(\nabla^\mu A^\nu) + c_2(\nabla_\mu A^\mu)^2 + c_3(\nabla_\mu A_\nu)(\nabla^\nu A^\mu). \tag{16}$$

The dual-channel specification fixes the free function as $\mathcal{F}_{\mathrm{dual}}$ evaluated at $x = \sqrt{\mathcal{K}}$:

$$\mathcal{F}(\mathcal{K}) = \tfrac{1}{2}\mathcal{K} - \sqrt{\mathcal{K}} + \ln\!\left(1 + \sqrt{\mathcal{K}}\right). \quad [\mathbf{P}] \tag{17}$$

Its asymptotics are the required ones: $\mathcal{F} \to \frac{1}{2}\mathcal{K}$ as $\mathcal{K} \to \infty$ (standard kinetic term, Newtonian branch) and $\mathcal{F} \to \frac{1}{3}\mathcal{K}^{3/2}$ as $\mathcal{K} \to 0$ (deep-MOND scaling), with $\mathcal{F}''(\mathcal{K}) > 0$ throughout [**P**].

B. Background and the screening ratio

On an FLRW background with $A^\mu = (1, \mathbf{0})$ one has $\mathcal{K}_0 = \alpha H^2(a)$ with $\alpha \equiv 3(c_1 + 3c_2 + c_3)$. The Friedmann equations are unmodified [**?**]: at the background level the theory is degenerate with ΛCDM [**P**]. The dimensionless background acceleration ratio today is

$$x_0 = \sqrt{\alpha}\,\frac{cH_0}{a_0} \approx 5.67\sqrt{\alpha}, \tag{18}$$

so for natural $\alpha = \mathcal{O}(1)$ the cosmological background sits on the *Newtonian* branch, $\mu(x_0) \approx 0.85$ — not in the deep-MOND regime. This is the origin of the screening. Expanding the effective gravitational coupling for linear perturbations,

$$\frac{G_{\rm eff}}{G} = 1 + \frac{\mathcal{F}''(\mathcal{K}_0)}{\mathcal{F}'(\mathcal{K}_0)}\,\delta\mathcal{K} + \mathcal{O}(\delta\mathcal{K}^2), \qquad \frac{\mathcal{F}''(\mathcal{K}_0)}{\mathcal{F}'(\mathcal{K}_0)} \approx 0.004. \quad [\mathbf{P}] \tag{19}$$

The MOND enhancement of linear growth is thus $\sim 0.4\%$, not the order-unity factor that a naive uniform-MOND treatment would give.

XI. SOLAR SYSTEM AND PPN LIMITS (D3)

A. The inner solar system is Newtonian by construction

At Earth's orbit the dimensionless MOND correction is $a_0/g \approx 2 \times 10^{-8}$, roughly $1137\times$ below the Cassini sensitivity $|\gamma-1| < 2.3\times10^{-5}$ [**D**]. This holds for *any* MOND interpolation function, since all of them satisfy $\mu \approx 1$ deep in the Newtonian regime; it is therefore not a test that discriminates $\mu = x/(1+x)$ from alternatives.

B. The external field effect and the Q_2 constraint

The discriminating solar-system observable is the quadrupolar distortion Q_2 induced by the external Galactic field ($g_{\rm ext} \approx 1.9 \times 10^{-10}\,\mathrm{m\,s^{-2}} \approx 1.6\,a_0$ at the Sun, i.e. squarely in the MOND transition regime). Park et al. [**?**], using the dataset behind the DE440 ephemerides, report

$$Q_2 = (1.6 \pm 1.8) \times 10^{-27}\,\mathrm{s}^{-2}, \tag{20}$$

consistent with zero, giving a bound $|Q_2| < 3.4\times10^{-27}\,\mathrm{s}^{-2}$. For classical AQUAL/QUMOND this is in 3–15σ tension with the external field effect implied by galactic rotation curves [**C**]. **The non-relativistic dual-channel action does not resolve this tension**; the resolution must come from the covariant completion.

C. Vainshtein screening

The Sun's MOND radius is $r_{\rm MOND} = \sqrt{GM_\odot/a_0} \approx 7031\,\mathrm{AU}$, so Earth orbits at $1.4 \times 10^{-4}\,r_{\rm MOND}$. Deep inside this radius the nonlinearity of $\mathcal{F}_{\rm dual}$ supplies Vainshtein-type screening [**?**] of the response to external perturbations, with suppression scaling as

$(r/r_{\rm MOND})^{3/2} \approx 1.7 \times 10^{-6}$. Combining the background factor $\mathcal{F}''/\mathcal{F}'$ with the Vainshtein factor gives

$$Q_2^{\rm RMOND} \approx 4.9 \times 10^{-29}\,{\rm s}^{-2}, \tag{21}$$

a factor ~ 70 below the Cassini bound [**D**], for *natural* $\mathcal{O}(1)$ couplings c_1, c_2, c_3. No fine-tuning is required. We emphasise the epistemic status: the $(r/r_{\rm MOND})^{3/2}$ scaling is standard, but this is a screening *estimate*, not a solution of the RMOND field equations with galactic boundary conditions [**O**]. The $70\times$ margin is what makes the conclusion robust to the estimate's uncertainty.

D. A falsifiable prediction

The framework predicts a residual quadrupole of order

$$Q_2 \sim 10^{-29}\,{\rm s}^{-2}, \tag{22}$$

roughly two orders of magnitude below present sensitivity. This is a genuine prediction rather than a consistency check: it is a nonzero value, it is specific, and it is within reach of next-generation ranging and space-based tests. A measurement establishing $|Q_2| \lesssim 10^{-30}\,{\rm s}^{-2}$ would place the theory under serious pressure.

Open [O]: exact PPN γ and β for specified c_1, c_2, c_3 require the post-Newtonian expansion of the RMOND metric. The framework is in place; the expansion has not been carried out.

XII. COSMOLOGICAL SECTOR (D5)

A. Why uniform MOND enhancement fails

Linear density perturbations obey $\ddot{\delta} + 2H\dot{\delta} = 4\pi G\rho_{\rm eff}\,\xi\,\delta$ with $\xi = G_{\rm eff}/G$. In an Einstein–de Sitter-like matter era the growing mode is $D \propto a^{\lambda}$ with

$$\lambda = \frac{-1/2 + \sqrt{1/4 + 6\xi}}{2}, \tag{23}$$

so $\xi = 1$ gives $\lambda = 1$ (ΛCDM) while a uniform $\xi = 2$ gives $\lambda = 1.366$ and $D(z = 0)$ larger by a factor ~ 76 [**D**]. Non-linear collapse amplifies this further. A universally applied MOND enhancement therefore *catastrophically* overproduces structure — consistent with the νHDM difficulties reported by Russell et al. [**?**].

B. Resolution by scale-dependent screening

The RMOND embedding of §X does not apply MOND universally. Because the cosmological background sits on the Newtonian branch, the linear enhancement is $\xi \approx 1.004$ rather than $\xi = 2$: the overproduction is suppressed by a factor ~ 250, from $76\times$ to 0.4% [**P**]. A sub-percent modification of linear growth is compatible with CMB and large-scale-structure constraints, which is the structural reason RMOND reproduces the acoustic peaks where non-relativistic MOND cosmologies do not.

Limits of this result [O]

Three limits are stated explicitly. (i) The 0.4% figure is *linear* theory; the non-linear regime requires N-body simulation in the Thomas et al. [**?**] formulation and has not been run. (ii) The linear CMB agreement is inherited from [**?**]; it has *not* been recomputed with $\mathcal{F}_{\text{dual}}$ in a modified Boltzmann code. (iii) Cluster scales remain untested (O5). We do not claim a complete cosmology — we claim that the specific overproduction obstruction is removed at linear order.

XIII. RELATIVISTIC STABILITY (D6)

The RMOND completion with $\mathcal{F}_{\text{dual}}$ satisfies the standard stability criteria [**P**]:

- **No Ostrogradsky ghosts:** $\mathcal{F}''(\mathcal{K}) > 0$ for all $\mathcal{K} > 0$, so the kinetic matrix is positive-definite.
- **Hamiltonian bounded below:** $\mathcal{F} - \frac{1}{2}\mathcal{K}\mathcal{F}' > 0$ on the physical branch.
- **Tensor sector unmodified:** $c_T = c$, satisfying GW170817 identically (§IX).
- **Strong coupling scale:** $\sim 10^{-10}\,\text{eV}$, far below any regime probed by laboratory or solar-system experiment.
- **Couplings:** natural positive c_1, c_2, c_3 satisfy all of the above simultaneously.

Open [O]: vector-sector perturbations may propagate superluminally on some backgrounds. This is a known feature of Einstein-aether-type theories and does not by itself imply a causality violation (no closed causal curves are exhibited), but a full characteristic analysis has not been performed.

XIV. LEAN 4 FORMAL VERIFICATION: SCOPE AND LIMITS

The Lean 4 suite (`05_lean_formalization/`) comprises **18 modules**, 0 `sorry`, axioms {`propext`, `Classical.choice`, `Quot.sound`}, Lean `v4.33.0-rc1`, Mathlib pinned at `5eec30bc`. Gate: `verify_all_proofs.sh`. The directory also contains Lean modules belonging to adjacent projects that share this package and its Mathlib pin; they are built and gated identically but are not results of this paper and are not enumerated in Table **??**. They are declared in `05_lean_formalization/ADJACENT_MODULES.txt` so the scoping is auditable.

Important provenance note: O6 — walked once in a clean worktree at 07185a6 (`lake exe cache get` + 17/17 PASS of the 17 targets then declared, `VERIFICATION_RUN_007`). Not yet demonstrated on a cold machine with empty host cache, and not yet a CI release gate. Artifact: `VERIFICATION_RUN_007/01_lean/` (transcript, per-target exit codes, toolchain and Mathlib revisions, SHA-256 checksums). The declared target inventory has since grown to 30; the gate was rerun on 2026-09-04 at 30/30, exit status 0, against a Mathlib checkout already present in `.lake/packages`, which is a warm-cache run and does not close O6.

XV. OPEN PROBLEMS

The following items are quarantined [**O**] in `OPEN_PROBLEMS_AND_TESTS.md`:

1. **O1 — a_0–Horizon Identity:** KMS thermal equilibrium formally yields $a = cH_0$ ($\xi = 1$) [**P**] (`HorizonScale.lean`). The $1/(2\pi)$ divisor remains unexplained [**O**]. Preregistered $a_0(z)$ test favours constant a_0 over horizon evolution at 5.9σ [**D**] (O4). The horizon-tied interpretation is disfavoured but not definitively killed pending larger samples.

2. **O2 — $\Omega_\Lambda = \ln 2$:** An asymptotic entropy boundary condition, not a dynamical prediction. A full curved-spacetime covariant derivation has not been written.

Module	What is proved [**P**] / What is not [**O**]
`AXIOMS_V2.lean`	*Assumption declarations.* States the five physical axioms A1–A5 as Lean typeclasses. Its two theorems are a field projection and the positivity of $\sqrt{g_{\mathrm{bar}} a_0}$; neither derives an axiom [**O**].
`CosmologicalSector.lean`	$0 < \ln 2 < 1$ and matter-density bounds [**P**]. $\Omega_\Lambda + \Omega_m = 1$ holds by construction, Ω_m being defined as $1 - \Omega_\Lambda$; it is not a flatness result. $\Omega_\Lambda = \ln 2$ is a definition, not a derivation [**O**].
`CovariantCompletion.lean`	RAQUAL superluminality; $\gamma_{\mathrm{PPN}} = 1$ (`disformal_gamma_ppn_unity`); preferred-frame parameters zero; FLRW decoupling $\Rightarrow \rho_\phi = 0$ [**P**], conditional on the stated embedding.
`DeSitterExtremal.lean`	Horizon lapse vanishing at $r = 1/H$; flat limit; $cH/(2\pi) > 0$ [**P**]. Does not derive $a_0 = cH_0/(2\pi)$ [**O**].
`DualChannelDerivation.lean`	Channel-balance identity $x - \frac{x}{1+x} = \frac{x^2}{1+x}$, i.e. $\mathcal{F}'_{\mathrm{dual}}(x) = x\,\mu(x)$; bounds $\mu(x) < x$ and $\mu(x) < 1$ [**P**]. Does *not* symbolically differentiate $\mathcal{F}_{\mathrm{dual}}$; physical mechanism of channel balance [**O**].
`GODActionKinematics.lean`	Dual-channel polynomial identity; AQUAL simple-μ ratio; BTFR algebraic scaling [**P**].
`HorizonScale.lean`	KMS thermal equilibrium formally yields $a = cH_0$ ($\xi = 1$); 2π cancels identically [**P**]. The $1/(2\pi)$ divisor is an unproved auxiliary postulate [**O**].
`ITActionClosure.lean`	Polynomial τ-law / simple-μ relation; deep-MOND BTFR; flat rotation curve for $n = 2$ [**P**]. Relativistic stability [**O**].
`MuProjection.lean`	Properties of $\mu(x)$; root uniqueness; power-law solves the dilaton EOM; exponential profile falsified [**P**]. Physical derivation of closure [**O**].
`PPNLimits.lean`	Solar-system PPN precision bounds at $x \geq 10^4$; Cassini radar delay satisfied at $x \geq 6 \times 10^7$ [**P**]. No asymptotic limit is formalized [**O**].
`PrintAxioms.lean`	*Diagnostic helper, not a physics module.* Emits `#print axioms` for the `CovariantCompletion` results. No proof content.
`PrintAxiomsD8.lean`	*Diagnostic helper, not a physics module.* Re-declares a subset of `TensorSpeed` in the same namespace and prints its axiom footprint; not an independent result.
`RelativisticStability.lean`	Kinetic convexity $\mathcal{F}'' > 0$ [**P**].
`SOCasimirGenuine.lean`	$\mathfrak{so}(n)$ Casimir eigenvalue from explicit generators [**P**]. Physical gauge symmetry [**O**].

3. **O3 — $\mu(x)$ Physical Mechanism:** The algebraic closure is proved; the physical origin of the dual-channel balance (bulk kinetic flux vs. horizon dissipation) is a motivated conjecture.

4. **O4 — $a_0(z)$ Redshift Test: Completed** [**D**]. Pre-registered test on 20 MUSE-DARK III galaxies favours $H_{\rm const}$ over $H_{\rm horizon}$ at 5.9σ. Extension to JWST/NIRSpec high-z samples pending. (Note: O4 in the original numbering referred to CMB/structure growth; the redshift test was added as a pre-registered protocol and has been completed. The CMB/structure-growth frontier remains open.)

5. **O5 — Cluster Lensing and Bullet Cluster:** MOND-family theories require additional physics for galaxy cluster mass budgets; applicability of this framework at cluster scales is untested.

6. **O7 — Necessity of the Padé Constraint (D2):** The uniqueness theorem of §V is conditional on Padé [1/1] minimality, which is assumed rather than derived. Whether it follows from a variational principle on the information geometry is open.

7. **O8 — Non-Linear Structure Formation (D5):** The 0.4% linear result of §XII does not extend to the non-linear regime without N-body simulation in the RMOND formulation [**?**]. Not run.

8. **O9 — Exact PPN Parameters (D3):** γ and β for specified c_1, c_2, c_3 require the post-Newtonian expansion of the RMOND metric. Framework complete; expansion not performed.

9. **O6 — Clean-Worktree Lean Build:** Walked once in a clean worktree at 07185a6 (`lake exe cache get` + 17/17 PASS, `VERIFICATION_RUN_007`). Not yet demonstrated on a cold machine with empty host cache, and not yet a CI release gate.

XVI. COSMOLOGICAL AND OPTICAL EXTENSIONS (OPEN RESEARCH PROGRAM)

The observer and cosmological sectors incorporate proposed phenomenological relations, all quarantined [**O**]:

- **Disformal Optical Metric:** $g_{\mu\nu}^{\rm opt} = g_{\mu\nu} + \beta(\chi)\nabla_\mu\chi\nabla_\nu\chi$.
- **Spectral Redshift Jacobian:** $1 + z_{\rm obs} = (1 + z_{\rm cosm})\mathcal{J}(\chi)$.
- **Luminosity Distance Modulation:** $d_L^{\rm eff}(z) = d_L(z)\sqrt{1 - \chi/\chi_{\rm crit}}$.
- **Horizon Density:** $\Omega_\Lambda(z = 0) = \ln 2$.

Full derivations from Maxwell's equations on curved spacetime and confrontation with CMB/BAO remain open (O4).

XVII. LIMITATIONS AND FALSIFIABILITY

The research program is subject to clear falsification criteria:

1. **CMB Power Spectrum:** If the Skordis–Złośnik completion with $\mathcal{J} = \mathcal{F}_{\rm dual}(\sqrt{\mathcal{Y}})$ cannot reproduce the Planck CMB acoustic peak positions and amplitudes, the covariant embedding is falsified as a complete cosmological theory.
2. **Structure Growth:** If the predicted matter power spectrum $P(k)$ is inconsistent with BOSS/eBOSS galaxy clustering, the model fails at cosmological scales.
3. **Cluster Mass Budgets:** Systematic failure on galaxy cluster rotation/lensing would restrict the framework to galactic scales only.
4. $a_0(z)$ **Redshift Test: Executed.** The pre-registered test favours constant a_0 over horizon evolution at 5.9σ [**D**]. The horizon-tied $a_0(z) = \xi cH(z)$ is excluded at $> 3\sigma$ on the current sample. This does not confirm the $1/(2\pi)$ normalization — it disfavours the horizon interpretation of a_0.
5. **Lean Build Failure:** Failure of `verify_all_proofs.sh` on a clean clone would retract all [**P**] claims pending repair.

XVIII. CONCLUSION

The Res-Nova v1.6.0 assessment establishes the following proved and computed results: the dual-channel action $\mathcal{F}_{\rm dual}(x)$ yields $\mu(x) = x/(1 + x)$ with exact correspondence limits

[**P**]; the single-channel arcsinh action is falsified [**P**]; pure RAQUAL and the disformal completion are falsified [**P**]; the dual-channel action embeds exactly into Skordis–Złośnik with $c_T = c$, $\gamma_{\rm PPN} = 1$, ghost freedom, and FLRW decoupling [**P**]; and the a_0 measurement from 171 SPARC galaxies is $1.116 \times 10^{-10} \pm 14.4\%$, consistent with the horizon anchor at 0.46σ [**D**].

To these v1.6.0 adds: conditional uniqueness of $\mathcal{F}_{\rm dual}$ under Padé minimality, with the odds-ratio and Fisher identities [**P**]; the RMOND completion with $\mathcal{F}(\mathcal{K}) = \mathcal{F}_{\rm dual}(\sqrt{\mathcal{K}})$, whose screening ratio $\mathcal{F}''/\mathcal{F}' \approx 0.004$ reduces the $76\times$ linear structure overproduction to 0.4% [**P**]; solar-system safety with $Q_2 \approx 4.9 \times 10^{-29}\,\mathrm{s}^{-2}$, a factor 70 below the Cassini bound at natural couplings [**D**]; and ghost freedom with a bounded Hamiltonian [**P**]. The theory's sharpest falsifiable prediction is the residual quadrupole $Q_2 \sim 10^{-29}\,\mathrm{s}^{-2}$.

The remaining open problems (O1–O9) are documented, bounded, and do not retract the proved core. They are of three kinds, and we distinguish them deliberately: *computational* (non-linear N-body, exact post-Newtonian expansion, CI automation), *observational* (larger high-z kinematic samples, next-generation ranging), and *structural* (the necessity of the Padé constraint, the $1/(2\pi)$ horizon normalization). None is a demonstrated inconsistency of the framework. The pre-registered $a_0(z)$ redshift test (O4) has been completed: constant a_0 is favoured over the horizon-tied $a_0(z) = \xi c H(z)$ at 5.9σ on 20 MUSE-DARK III galaxies [**D**]. This disfavours the horizon interpretation of a_0 but does not falsify the dual-channel framework itself, which stands on the proved variational closure, covariant embedding, and SPARC benchmark. The primary scientific frontier is now CMB and structure-growth consistency on FLRW backgrounds. This is the next derivation target in the Res-Nova research program.

DATA AND CODE AVAILABILITY

All raw data, Python scripts, Lean 4 modules, and verification manifests are openly archived in the public repository at `https://github.com/Mega-Therion/Res-Nova` (release `v1.6.2`), with DOI 10.5281/zenodo.21969121; verified artifact references `VERIFICATION_RUN_003`, `VERIFICATION_RUN_005`, `VERIFICATION_RUN_006`, and `VERIFICATION_RUN_007`. A permanent snapshot with a DOI is deposited on Zenodo.

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APPENDIX A: LEAN 4 SOURCE INVENTORY AND VERIFICATION PROVENANCE

The tracked Lean 4 sources under `05_lean_formalization/` comprise **18 modules**, enumerated in Table II. This appendix records the headline declarations per module and the provenance of the verification run; it adds no claim beyond that table.

a. Verification provenance. O6 — walked once in a clean worktree at 07185a6 (`lake exe cache get` + 17/17 PASS of the 17 targets then declared, `VERIFICATION_RUN_007`). Not yet demonstrated on a cold machine with empty host cache, and not yet a CI release gate. At that commit the gate `verify_all_proofs.sh` elaborated all 17 declared targets with exit status 0, no `sorry` in proof code, and axiom footprints confined to {`propext`, `Classical.choice`, `Quot.sound`}, under Lean `v4.33.0-rc1` and Mathlib `5eec30bc`. The transcript, per-target exit codes, toolchain and Mathlib revisions, and SHA-256 checksums are archived in `VERIFICATION_RUN_007/01_lean/`.

b. What elaboration does and does not certify. A green gate run certifies that each statement follows from its encoded definitions. It does not certify that those definitions model the physical system, and it does not certify assumptions carried as typeclass fields, structure fields, or theorem hypotheses. The suite declares no Lean global `axiom`; that is a statement about `axiom` declarations only. See `THEORY_ASSUMPTION_AUDIT.md`.

c. Scope note on `SovereignRegularity.lean`. The module bounds pointwise vorticity by an assumed control field. It does not formalize an integral and does not formalize a partial differential equation, so it does not establish the Beale–Kato–Majda criterion [**?**], which concerns $\int_0^T \|\omega\|_\infty \, dt$. `sovereign_regularity_theorem` restates the module's own hypothesis.

APPENDIX B: SPARC METRIC DEFINITIONS

For each galaxy g with N_g observed kinematic radii R_i, observed velocities $V_{\mathrm{obs},i}$, and observational uncertainties $\sigma_{V,i}$, the model velocity is:

$$V_{\mathrm{model}}(R_i) = V_{\mathrm{bar}}(R_i) \cdot \left[\frac{1}{2} + \sqrt{\frac{1}{4} + \frac{a_0}{g_{\mathrm{bar}}(R_i)}} \right]^{1/2}, \tag{24}$$

where $g_{\mathrm{bar}}(R_i) = V_{\mathrm{bar}}(R_i)^2/R_i$, with $V_{\mathrm{bar}}^2(R) = |V_{\mathrm{gas}}|V_{\mathrm{gas}} + \Upsilon_{\mathrm{disk}}|V_{\mathrm{disk}}|V_{\mathrm{disk}} + \Upsilon_{\mathrm{bulge}}|V_{\mathrm{bulge}}|V_{\mathrm{bulge}}$.

Module	Headline declarations
Proof modules	
CosmologicalSector.lean	log_two_pos, log_two_lt_one, matter_density_bounds, spatial_flatness_sum
CovariantCompletion.lean	raqual_superluminal_obstruction, disformal_gamma_ppn_unity, preferred_frame_parameters_zero, flrw_projected_gradient_zero, no_dynamical_dark_energy_density
DeSitterExtremal.lean	desitter_lapse_horizon, expr_cH_over_2pi_pos, desitter_flat_limit, volume_law_weight_nonneg
DualChannelDerivation.lean	dual_channel_flux_algebra, mu_derived_inversion, mu_derived_deep_mond_upper_bound, mu_derived_newtonian_bound
GODActionKinematics.lean	dual_channel_poly_identity, aqual_simple_mu_ratio, btfr_algebraic_scaling
HorizonScale.lean	kms_cancellation_equilibrium, horizon_acceleration_ratio_is_one, verlinde_entropic_cancellation
ITActionClosure.lean	tauLaw_eq_simple_mu_poly, btfr_deep_mond, enclosed_mass_n2_linear, flat_rotation_curve_n2
MuProjection.lean	mu_simple_eq_cos, quadratic_law_root_unique, powerLaw_solves_dilaton_eom, powerLaw_iterated_deriv, exp_profile_fails_cubic
PPNLimits.lean	fractional_deviation_eq, solar_system_precision_bound, cassini_radar_delay_satisfied
RelativisticStability.lean	first_derivative_pos, second_derivative_pos, ghost_free_convexity
SOCasimirGenuine.lean	gen_skew, sum_dg, casimir_defining_rep, gen_sq, casimir_scalar_eq
SkordisZlosnikEmbedding.lean	sz_aqual_reduction, dJ_dY_pos, sz_tensor_speed_luminal, sz_weak_field_lensing
SovereignRegularity.lean	chiral_iff_lipschitz_constant, lipschitz_implies_angle_modulus, bkm_no_blowup,

Per-galaxy metric

$$\frac{\chi^2_{\text{data}}}{N_g} = \frac{1}{N_g}\sum_{i=1}^{N_g}\left(\frac{V_{\text{obs},i} - f_d V_{\text{model}}(f_d R_i)}{\sigma_{V,i}}\right)^2. \tag{25}$$

These are definitions only. The sample size, parameter count, and all fitted values are carried by the frozen artifacts named in Appendix C; they are deliberately not restated here, so that this appendix cannot drift from them.

APPENDIX C: PARAMETER ACCOUNTING AND CROSS-VALIDATION

d. Superseded accounting — do not quote. Earlier revisions of this appendix reported an aggregate residual over $N_{\text{data}} = 3{,}391$ points with $N_{\text{par}} = 382$ and $\text{DOF}_{\text{nom}} = 3{,}009$, and a 5-fold cross-validation evaluated against *fixed* population priors. Both are superseded: items **D4.1** and **D4.2** of `EPISTEMIC_BOUNDARY_v1.5.0.md` record the method defects — radial points treated as independent, and a cross-validation that held a_0 fixed across folds and so was not a cross-validation. Those numbers are retained here only to be labelled, and must not be quoted as current results.

e. Current accounting. The parameter ledger, sample definition, systematic budget, and honest per-fold cross-validation are carried by the frozen artifacts on `main`:

- `02_galaxy_dynamics/A0_MEASUREMENT.json` — working a_0 with the statistical and systematic budget (item D4.3).
- `02_galaxy_dynamics/PARAMETER_LEDGER.json` — tiered parameter counts and per-tier medians (items D4.7, D4.8).
- `02_galaxy_dynamics/NFW_CONSTRAINED.json` — concentration-prior comparison (item D4.9).
- `02_galaxy_dynamics/A0_ESTIMATE.json` — cross-validation retraining a_0 per fold (item D4.6).
- `02_galaxy_dynamics/SPARC_PARAMETER_BUDGET.md` — the budget derivation.

Where this appendix and `EPISTEMIC_BOUNDARY_v1.5.0.md` appear to differ, the boundary document governs.

APPENDIX D: REPRODUCTION COMMANDS

All paths are relative to the repository root. SPARC rotation-curve tables are not vendored; see `02_galaxy_dynamics/SPARC_DATA.md` for the official source and the checksum manifest.

```
# 1. Lean 4 formal suite (all 17 declared targets)
cd 05_lean_formalization
lake exe cache get          # fresh-clone path (see 06)
./verify_all_proofs.sh      # exits 0 only if every target elaborates cleanly

# 2. SPARC measurement and parameter ledger
#    Requires the rotation-curve tables; see 02_galaxy_dynamics/SPARC_DATA.md
cd 02_galaxy_dynamics
python3 a0_measure.py
python3 parameter_ledger.py

# 3. Manuscript PDF
pdflatex -interaction=nonstopmode final_manuscript.tex
bibtex   final_manuscript
pdflatex -interaction=nonstopmode final_manuscript.tex
pdflatex -interaction=nonstopmode final_manuscript.tex
```

APPENDIX E: ARTIFACT MANIFESTS

Earlier revisions printed a table of truncated SHA-256 prefixes, which cannot be verified. The full manifests are tracked in the repository and should be checked directly:

- `VERIFICATION_RUN_001/02_sparc_strict_135/RAW_DATA_MANIFEST.sha256` — SHA-256 of each of the 175 `*_rotmod.dat` inputs. Verify with `sha256sum -c` from the data directory.
- `VERIFICATION_RUN_003/01_lean/SHA256SUMS.txt` — checksums of the Lean gate transcript and its machine-readable result record.

- `05_lean_formalization/lake-manifest.json` — pinned Mathlib revision and transitive dependency revisions.
- `05_lean_formalization/lean-toolchain` — pinned Lean toolchain.